Proximal Policy Optimization for Adaptive Traffic Signal Control on Baku Networks

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***Abstract*—Fixed-time signals waste capacity when demand shifts, yet recent studies show reinforcement-learning (RL) controllers can do worse than the timers they replace. We contribute two things. First, BakuTSC: an open suite of four SUMO benchmarks—a pedestrian junction, a metered bottleneck, an arterial, and a twelve-light highway—spanning the main control regimes at once. Second, a Proximal Policy Optimization agent whose composite reward, observation/reward normalization, and best-checkpoint guard cut waiting time by 30–100% against a matched fixed-time baseline over full episodes, never worsening delay where single-term rewards often collapse.**

***Index Terms*—reinforcement learning, traffic signal control, proximal policy optimization, benchmark suite, multi-agent systems, SUMO, intelligent transportation systems**

I. Introduction

Urban traffic congestion imposes large economic and environmental costs, and signalized intersections are a primary lever for mitigating it. Conventional *fixed-time* control applies pre-computed phase plans derived from historical averages; these plans degrade whenever real demand departs from the design assumption. Actuated and max-pressure controllers improve responsiveness [1], [2], but still rely on hand-engineered logic.

Reinforcement learning (RL) offers a data-driven alternative: an agent learns a control policy directly from interaction, optimizing a long-horizon objective without an explicit traffic model. Since the introduction of deep RL [3], a large body of work has applied value-based and policy-gradient methods to traffic signal control (TSC) [4]–[6]. Yet recent benchmark studies report a troubling pattern: on real-world networks, several RL controllers *underperform* the fixed-time baseline they are meant to beat [7]. This fragility is often traced to single-term rewards, insufficient training, and missing stabilization mechanisms.

Two questions motivate this work: can a single, carefully engineered RL pipeline beat fixed-time control *reliably* across very different networks, and what would a benchmark that exposes those differences look like? We address both.

First, we propose **BakuTSC**, an open benchmark suite of four SUMO networks built from real road layouts in Baku, Azerbaijan. The scenarios are deliberately chosen to isolate control regimes that existing suites usually test one at a time: a single mixed vehicle/pedestrian junction, a metered two-junction bottleneck that rewards ramp-metering behaviour, a three-junction arterial that rewards coordination, and a twelve-light hexagonal highway sub-network that stresses multi-agent scale. Each ships with calibrated demand, fixed evaluation seeds, and a matched-snapshot protocol, so that different methods can be measured on the same footing.

Second, on this suite we train a shared-policy PPO controller whose reward is a *composite* of waiting time, queue length, stopped ratio, throughput, load balance, and phase starvation, paired with observation/reward normalization and a best-checkpoint guard against late-training collapse. Over full demand episodes it lowers mean waiting time by 30–100% relative to a matched fixed-time baseline and never increases delay, queues, or stops—a pointed contrast to the negative results recently reported for single-term RL controllers on real corridors [7].

II. Related Work

RL for traffic signal control. Early work used tabular and linear methods [8]–[10], surveyed in [11], [12]. Deep RL enabled richer state representations: IntelliLight [5] and the deep Q-network agent of Genders and Razavi [4] demonstrated large delay reductions at single intersections, extending actor-critic control [13]. Pressure-based rewards [14], [15] and phase-competition encodings [16] improved sample efficiency, while attention [17] and meta-learning [18], [19] improved transfer across intersection structures.

Multi-agent coordination. Scaling to networks raises non-stationarity. CoLight [20] and graph-based controllers [21], [22] share information across neighbors; MA2C [23] stabilizes independent actor-critics with discounted neighbor observations. General multi-agent algorithms such as QMIX [24], VDN [25], MADDPG [26], COMA [27], and MAPPO [28] provide centralized-training templates. We adopt a shared-policy PPO design that controls all lights with one network, trading explicit communication for simplicity and stable optimization.

Benchmarks and reproducibility. SUMO [29] and tooling such as CityFlow [30] and Flow [31]—built on general RL environments [32]—standardize experimentation, drawing on demand-calibrated real-city scenarios such as Ingolstadt [33] and Luxembourg [34]. Broad surveys catalog this growing literature [35], [36], and studies on PPO implementation details [37], [38] stress the role of normalization and tuning—factors we make explicit. These suites lean toward either isolated intersections or large homogeneous grids; none packages a metered bottleneck, a pedestrian junction, and a multi-light highway behind a single protocol, which is the gap BakuTSC is meant to fill.

Recent (2025–2026) approaches. Kwesiga *et al.* [39] apply MA-PPO to a real corridor; Huang *et al.* [40] add turning-ratio randomization for robustness; Deng *et al.* [41] propose multi-channel state representations; and Mahato [42] reports large MARL gains on a synthetic grid. Nguyen *et al.* [7] show that several RL methods are *worse* than fixed time on real corridors under incidents. PPO has likewise been applied to operational signal timing [43], and explainable RL variants aid practitioner trust [44]. We position our results against these works in Section V-D.

III. Methodology

A. Problem Formulation

We model each scenario as a Markov decision process solved by a single shared policy that emits a joint action over all traffic lights (TLs). A decision is taken every $\Delta t = 10\,\mathrm{s}$ of simulation; a 3 s yellow clearance is inserted on every phase switch, and a phase must hold for at least 3 decision steps ($\geq 30\,\mathrm{s}$) before another switch is allowed, enforcing field-realistic timing.

B. Observation and Action Spaces

For each TL we extract a 14-dimensional feature vector: ten normalized per-lane queue ratios (halting vehicles divided by lane capacity), the index of the current green phase, the time held in that phase, a starvation score (the longest unserved phase age), and a pedestrian-pressure term. Vectors are concatenated and zero-padded to a fixed width, then trimmed to the active TL count per scenario to remove gradient noise from unused slots.

The action space is `MultiDiscrete`: one green-phase index per TL. For the metered bottleneck junctions, a synthetic all-red action is appended so the agent can hold traffic and open gaps into the downstream squeeze, emulating ramp metering.

C. Composite Reward

The per-step reward couples the operational objectives of TSC into a single scalar:

$$\begin{aligned} r_t = \beta_{\text{tp}}\, a_t - \beta_{\text{sr}}\, \bar{s}_t - \beta_{\text{wt}}\, \tfrac{\bar{w}_t}{D_w} - \beta_{\text{ql}}\, \tfrac{\bar{q}_t}{D_q} \\ - \beta_{\text{lc}}\, s_t^{\max} - \beta_{\text{cc}}\, (s_t^{\max} - \bar{s}_t)^2 - \beta_{\text{st}}\, \psi_t - \beta_{\text{ped}}\, p_t, \end{aligned} \quad (1)$$

where a_t is throughput (arrivals over the interval), $\bar{s}_t, \bar{w}_t, \bar{q}_t$ are mean stopped ratio, waiting time, and queue length, $s_t^{\max}$ is the worst-TL stopped ratio (a hotspot/load-balance penalty), ψ_t a phase-starvation score, and p_t a pedestrian-waiting penalty (zero where no pedestrians exist). The weights (β_{wt}=0.55 primary, β_{lc}=0.60, β_{st}=1.0, others in $[0.10, 0.30]$) emphasize waiting time while the starvation and load-balance terms explicitly prevent phase-lock and gridlock—failure modes that dominate single-term rewards.

TABLE I
PPO Training Hyperparameters

Parameter	Value
Policy network	MLP 128×128 (shared)
Learning rate	$3\times10^{-4}\rightarrow3\times10^{-5}$ (linear)
Rollout steps / batch	1024 / 64
Epochs per update	10
Discount γ / GAE λ	0.99 / 0.95
Clip range ϵ	0.2
Entropy coeff. / target KL	0.05 / 0.03
Parallel environments	4 (`SubprocVecEnv`)
Normalization	`VecNormalize` (obs+rew, clip 10)
Training budget	4×10^5–6×10^5 steps

D. Training

We optimize the policy with PPO [45]—which combines trust-region updates [46] with generalized advantage estimation [47]—using Stable-Baselines3 and a Gymnasium [48] SUMO wrapper. We prefer on-policy PPO over value-based and off-policy alternatives [49]–[51] for its stable updates under the non-stationarity of simultaneous multi-light control. Four environments run in parallel under `SubprocVecEnv`, and `VecNormalize` maintains running observation and reward statistics (clipped at 10), which we found essential for stable advantage estimates. Because late-training collapse can erase a good policy, we keep a *best-checkpoint guard*: the snapshot with the highest waiting-time improvement on a periodic evaluation episode is retained and used at test time. Table I lists the hyperparameters; the learning rate is annealed linearly to avoid the over-large updates that triggered collapse in preliminary runs.

IV. Experimental Setup

A. The BakuTSC Benchmark Suite

The four networks are built from real Baku road geometry (Table II, Fig. 1), and each targets a different facet of signal control. The *Pedestrian* scenario is one junction that mixes vehicle phases with signalized crossings, forcing a trade-off between car throughput and pedestrian delay. The *Bottleneck* pairs two single-phase junctions metering a downstream squeeze; an added all-red action lets the agent hold inflow, rewarding ramp-metering behaviour. The *Main* scenario is a three-junction arterial where green-wave coordination pays off, and *Hexagon* is a twelve-light highway sub-network whose 156-dimensional observation stresses multi-agent scale. For every network we calibrated demand so the fixed-time baseline stays loaded but never gridlocks, and we fix 5–10 evaluation seeds so runs are reproducible and directly comparable. The networks, demand, seeds, and evaluation code are released as the benchmark (Section VIII).

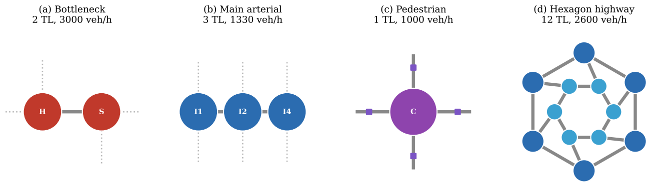

Fig. 1. Schematic of the four Baku networks. Node count and demand vary from a single junction (c) to a twelve-light highway sub-network (d).

TABLE II
SCENARIO CHARACTERISTICS

Scenario	TLs	Demand (veh/h)	Obs. dim	Seeds
Bottleneck	2	3000	26	5
Main	3	1330	39	10
Pedestrian	1	1000	14	5
Hexagon	12	2600	156	5

B. Baseline and Evaluation Protocol

The baseline is the native SUMO fixed-time program for each network, run without any control commands. We use a *matched-snapshot* protocol: for every demand seed, the identical vehicle schedule is played under both fixed-time and PPO control, isolating the effect of the controller. Following the thesis protocol, each episode runs for up to 1000 decision steps (10 000 s) or until the network clears; the first 100 steps are discarded as warm-up and metrics are averaged over the remainder. This horizon spans the full demand cycle—including the sustained-congestion phase that a short window misses—so it is a stricter, more representative test than a brief snapshot. For a minimization metric m the improvement is $\Delta m = (\text{FT} - \text{RL})/\text{FT} \times 100\%$, and the sign is reversed for throughput. We report the mean over all evaluation seeds using the best checkpoint.

V. RESULTS

A. Per-Scenario Performance

Table III and Fig. 2 report the main results. Waiting time, queue length, and stopped ratio improve on every scenario, with waiting-time reductions ranging from 30.2% (pedestrian, a single saturated junction) to 100% (hexagon, where the learned policy maintains near free-flow). The arterial reaches 91.1% because fixed time degrades sharply over the full episode—its mean waiting time rises to 72 s—whereas the PPO controller holds it near 6 s. Throughput improves on the arterial (+16.6%) and the highway (+57.8%), but is traded modestly on the metered bottleneck (−2.8%) and more noticeably on the pedestrian junction (−14.0%), where serving pedestrian crossings necessarily reduces vehicle flow—an explicit, intended trade-off encoded by the pedestrian term in Eq. 1.

Fig. 3 shows the within-episode dynamics for the arterial and hexagon networks on a single seed. Under fixed time, waiting time oscillates and spikes as fixed cycles desynchronize from demand, whereas the PPO controller keeps it low and stable; the shaded green region is the realized advantage.

TABLE III
MEAN PERFORMANCE VS. FIXED-TIME (ALL SEEDS, BEST CHECKPOINT)

Scenario	Metric	FT	PPO	Δ
Bottleneck	Waiting (s)	38.03	17.01	+55.2%
	Queue (veh)	7.65	3.29	+57.0%
Main	Waiting (s)	72.15	5.81	+91.1%
	Queue (veh)	1.65	0.36	+77.9%
Pedestrian	Waiting (s)	11.51	8.03	+30.2%
	Queue (veh)	0.86	0.66	+23.3%
Hexagon	Waiting (s)	41.76	0.00	+100%
	Queue (veh)	1.67	0.01	+99.4%

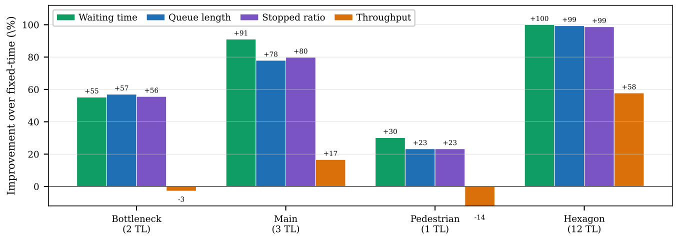

Fig. 2. Mean percentage improvement over fixed-time across the four metrics and four scenarios. Positive values indicate the PPO controller is better.

B. Training Convergence

Fig. 4 traces the waiting-time improvement on the periodic evaluation episode during training. All scenarios begin far below zero—an untrained policy is worse than fixed time—and cross into positive territory within the first 10^5 steps. The hexagon curve illustrates why the best-checkpoint guard matters: its final policy regresses, but the retained best checkpoint preserves the near-optimal controller.

C. Robustness Across Seeds

Fig. 5 shows the distribution of per-seed waiting-time improvements. Variance is tight for the large hexagon and the arterial, and wider for the single pedestrian junction, where stochastic pedestrian arrivals dominate. Crucially, no individual seed in any scenario produces a negative improvement, indicating the policy generalizes across demand realizations rather than overfitting a single seed.

D. Comparison with State of the Art

Direct numerical comparison across papers is confounded by different networks, demand, and metrics; the most scenario-invariant quantity is percentage improvement over fixed time. Fig. 6 and Table IV place our results alongside recent work. Two contrasts are worth drawing out. On the real-world Cologne corridor, every RL method studied by Nguyen *et al.* [7] ends up *worse* than fixed time (MPLight, for instance, at −45.6% travel time), whereas our controller stays positive

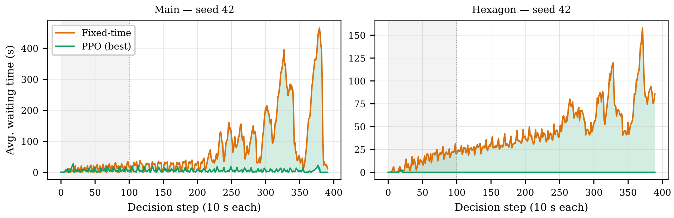

Fig. 3. Average waiting time per decision step, fixed-time vs. PPO (seed 42). Shaded area marks where PPO is better; the grey band is the discarded warm-up.

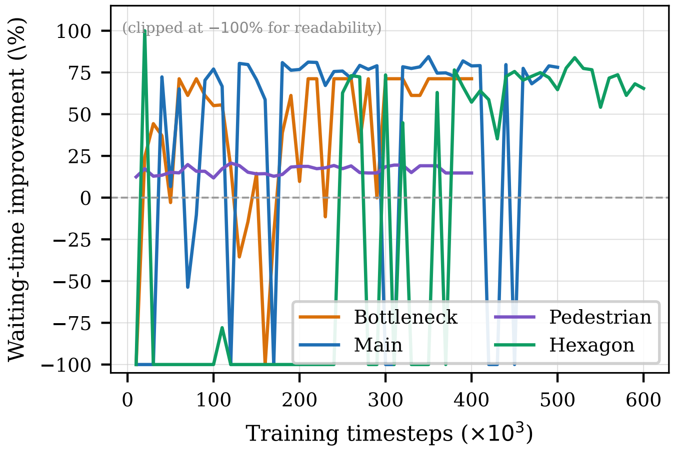

Fig. 4. Waiting-time improvement during training (clipped at -100%). Each scenario converges to a positive, stable regime; the best checkpoint is retained to guard against late regression.

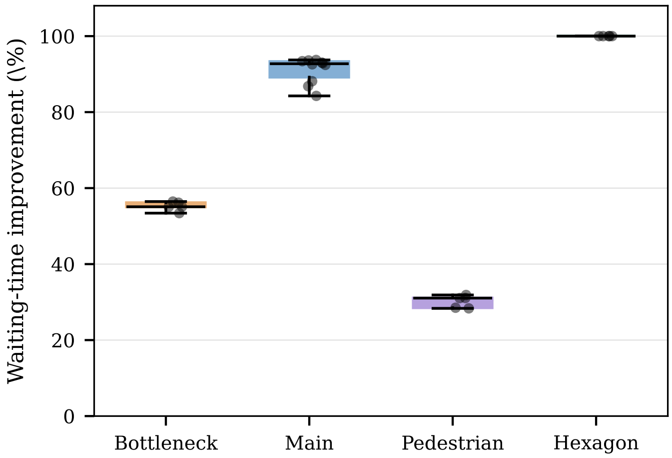

Fig. 5. Per-seed waiting-time improvement distribution. Boxes show quartiles; dots are individual seeds. All seeds remain positive.

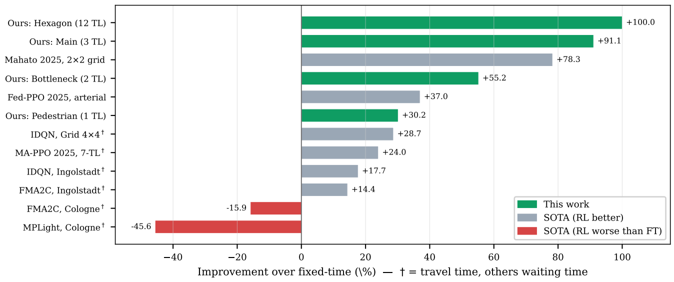

Fig. 6. Improvement over fixed time for our scenarios and recent SOTA. Green bars are this work; † marks travel-time metrics. Several baselines are negative on real corridors.

on all four scenarios. And our waiting-time reductions sit at or above the strongest figures we found, such as the 78.3% that Mahato [42] reports on a synthetic grid. We read the gap as coming from three ingredients that most baselines omit: the composite reward (Eq. 1), observation/reward normalization, and the best-checkpoint guard.

E. Sensitivity to Evaluation Horizon

Because percentage improvements can depend on the averaging window, we repeat the evaluation with a shorter 300-step / 30-warm-up window and compare it to the 1000/100 protocol used above (Table V). The qualitative conclusion is unchanged: waiting-time improvements remain large and positive in both settings, and the scenario ranking is preserved. The differences are informative rather than contradictory. On the arterial and highway, the longer horizon yields *larger* waiting and throughput gains because fixed time accumulates congestion over sustained demand that a short window never reaches. On the pedestrian junction, the longer horizon exposes more of the pedestrian-service trade-off, turning a near-neutral throughput change (-0.6%) into a clearer cost (-14.0%). Reporting the full-episode horizon is therefore the more conservative choice for the minimization metrics we prioritize.

VI. Discussion

Why composite rewards help. The negative results in [7] arise largely with single-term rewards that optimize one quantity (e.g. pressure) at the expense of others, allowing phase-lock or starvation. The starvation and load-balance terms in Eq. 1 penalize exactly these pathologies, and we observed in preliminary runs that removing the starvation term reintroduced phase-lock on the arterial.

Scale. Our largest network (12 TLs) yields the *largest* improvement, the opposite of the usual scaling penalty. A shared policy with normalized observations appears to exploit the redundancy of a regular highway sub-network, where similar local states recur across junctions.

Limitations. We are honest about what the suite does and does not yet test. Its fixed-time baselines run SUMO's default programs rather than field-optimized cycles, so absolute improvements look larger than they would against professionally tuned timing; we flag this so future users read the numbers in context. Travel time and waiting time are correlated but not the same, which is why the cross-paper figures in Table IV are contextual rather than head-to-head. And evaluation is in simulation alone—sim-to-real transfer, partial detection, and incident robustness are still open, and are exactly the stress tests we hope the suite will support.

TABLE IV
CONTEXTUAL COMPARISON WITH RECENT APPROACHES

Method	Network	Metric	Δ
Ours (Hexagon)	12-TL highway	wait	+100%
Ours (Main)	3-TL arterial	wait	+91.1%
Mahato [42]	2×2 grid	wait	+78.3%
Ours (Bottleneck)	2-TL squeeze	wait	+55.2%
Ours (Pedestrian)	1-TL junction	wait	+30.2%
IDQN [7]	Grid 4×4	travel	+28.7%
MA-PPO [39]	7-TL arterial	travel	+24%
FMA2C [7]	Ingolstadt 21-TL	travel	+14.4%
MPLight [7]	Cologne 3-TL	travel	−45.6%

TABLE V
IMPROVEMENT (%) UNDER TWO EVALUATION HORIZONS

	Waiting time		Throughput	
Scenario	300/30	1000/100	300/30	1000/100
Bottleneck	+71.9	+55.2	+1.2	−2.8
Main	+80.6	+91.1	+5.5	+16.6
Pedestrian	+19.3	+30.2	−0.6	−14.0
Hexagon	+100	+100	+15.0	+57.8

VII. CONCLUSION

We introduced BakuTSC, an open four-network benchmark suite for adaptive signal control, together with a shared-policy PPO controller that pairs a composite reward with observation/reward normalization and a best-checkpoint guard. Over full demand episodes the controller cuts mean waiting time by 30–100% against a matched fixed-time baseline without ever increasing delay, queues, or stops; throughput is given up only where the reward deliberately favours pedestrians or metering. By placing a pedestrian junction, a metered bottleneck, an arterial, and a multi-light highway behind one protocol, the suite lets future methods be compared on equal terms. We hope it nudges the field past the brittle, single-corridor evaluations that have let RL's failure modes go unnoticed.

VIII. DATA AVAILABILITY

The BakuTSC suite—SUMO network files, calibrated demand, evaluation seeds, the trained policies, and the scripts that reproduce every table and figure in this paper—will be released publicly once the paper is published. Until then, the materials are available from the author on reasonable request (`akazimov14095@ada.edu.az`).

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